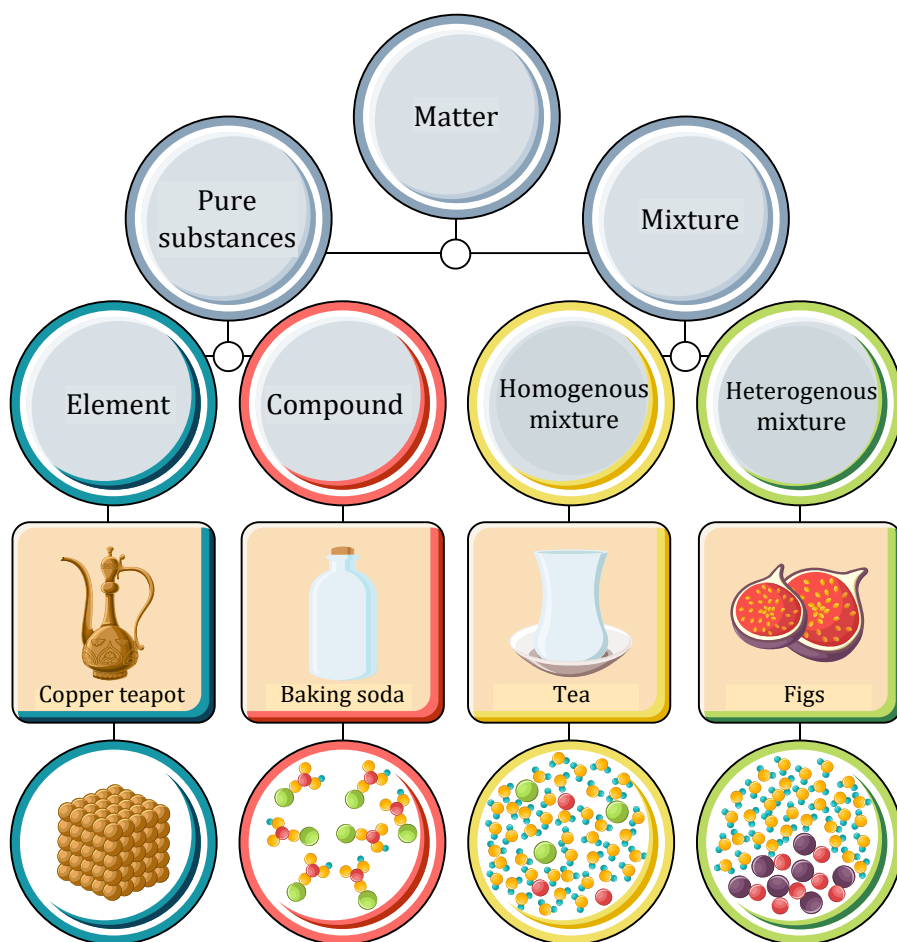


2

Is Matter Around us Pure



"Most of the matter around us exist as a mixture of two or more pure components."

1. Introduction

We know that all the matter around us is not pure. If we observe some soil and some sugar placed on two different sheets of paper with a magnifying glass, we find that soil contains clay particles, some grass particles and even some dead insects, etc. That is, soil contains particles of different kinds, thus it is called an impure substance (or mixture). Now observe sugar, it contains only one kind of particles, thus it is called a pure substance.

For a common man, pure actually means having no adulteration. But for a scientist pure means that it contains only one type of matter or particles all of which are same in their chemical nature.

SPOT LIGHT

A substance which has only one kind of component and nothing else in it is called a pure substance. But it is quite difficult to get pure substance. Substances are mostly mixed with one another and their combination is known as mixture.

2. Types of matter

On the basis of chemical nature, matter can be classified into two types:

- (i) Pure substance
- (ii) Impure substance (mixture)

Pure substance

A homogeneous material which contains particles of only one kind and has a definite set of properties, is called a pure substance. For example, O_2 , iron, sulphur contains only kind of particles, whereas sodium chloride is a pure substance, because it has a fixed number of sodium and chloride ions, combined together in fixed proportion by weight. Similarly, magnesium oxide (MgO), carbon dioxide (CO_2), copper sulphate ($CuSO_4$), etc. are pure substances which have definite set of properties.

Characteristics of pure substances

- (1) A pure substance is homogeneous in nature.
- (2) A pure substance has a definite set of properties.
- (3) The composition of a pure substance cannot be altered by any physical means.
- (4) A pure substance has a definite melting point, density, boiling point, etc.



Salt in water is an example of homogeneous mixture



Can we consider all homogeneous substances as pure?

Explanation

No, Common salt solution in water is a homogeneous solution. Yet it cannot be called a pure substance, as it is made up of two different substances i.e. salt and water.

Mixture

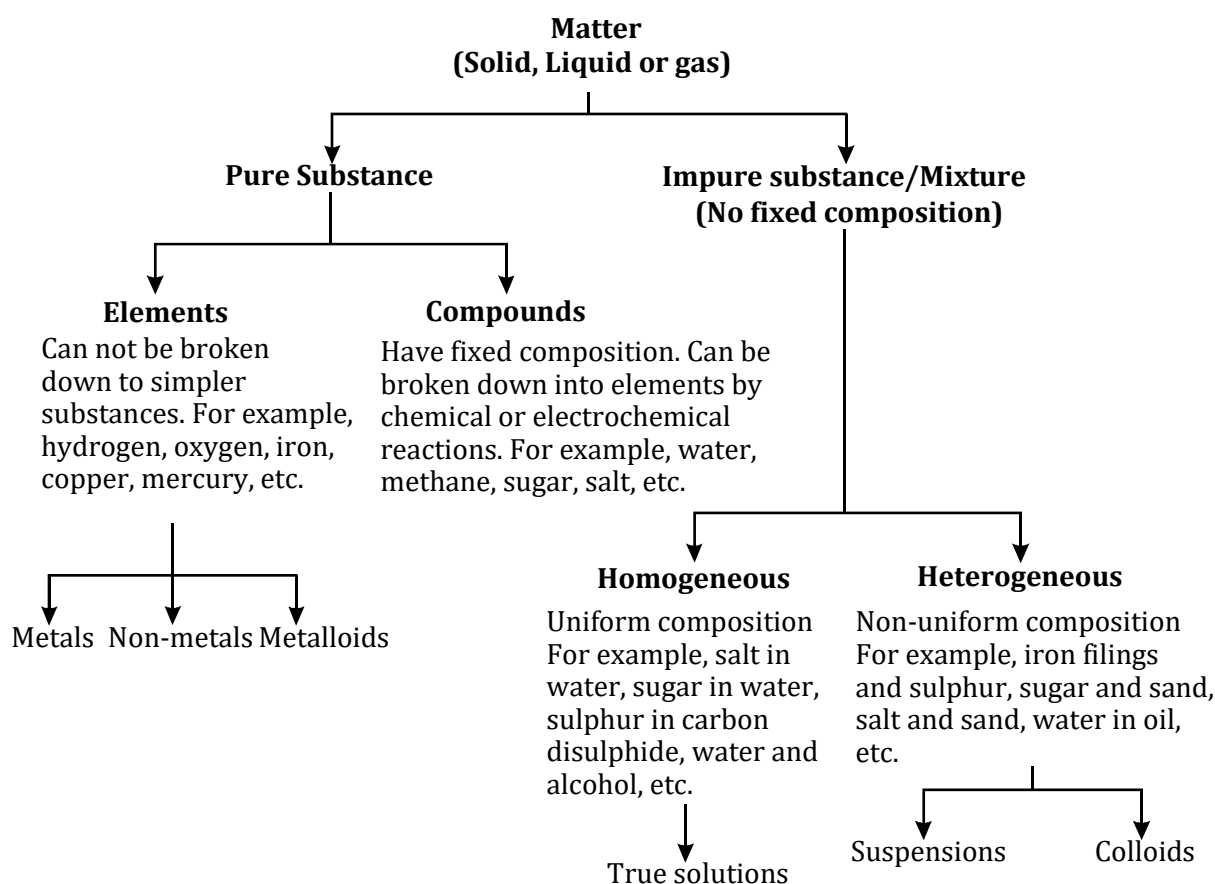
When two or more substances (elements, compounds or both) are physically mixed together in any proportion, such that they do not undergo any chemical change, but retain their individual characteristics, the resulting product is called a mixture.

For example, Brass is a mixture of copper and zinc.

Crude oil is a mixture of large number of different hydrocarbons. If we dissolve sugar in water both of which are pure compounds, the solution of sugar in water is a mixture.

Characteristics of mixtures

- (1) A mixture may be homogeneous or heterogeneous.
- (2) The constituents of a mixture can be separated by physical means like filtration, evaporation, sublimation and magnetic separation.
- (3) In the preparation of a mixture, energy is neither evolved nor absorbed.
- (4) A mixture has no definite melting point and boiling point.
- (5) The constituents of a mixture retain their original set of properties. For example, magnet attracts iron filings in a mixture of sand and iron filings.



3. Types of mixtures

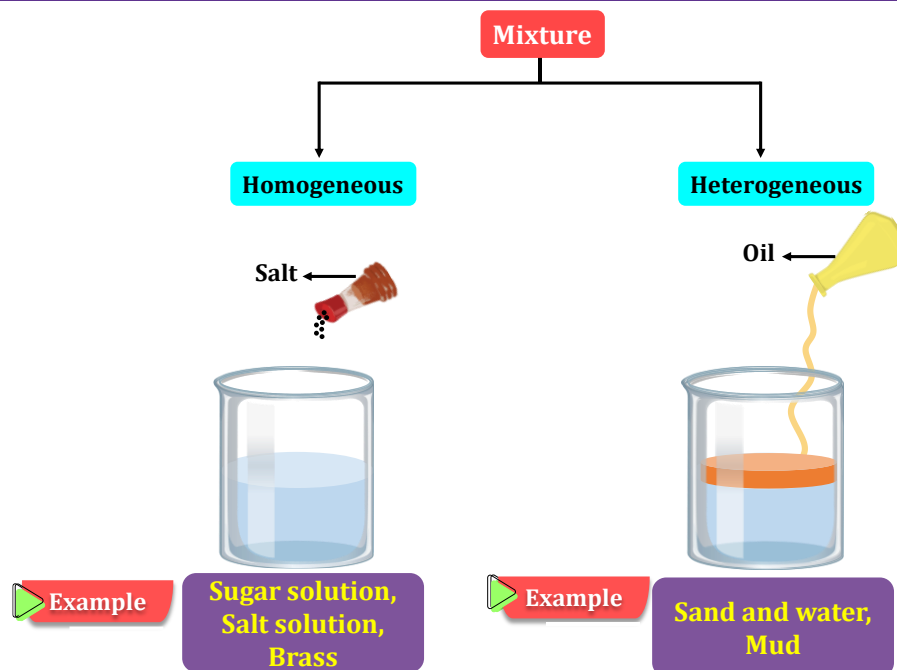
Depending upon the nature of components, a mixture can be divided into two types.

(a) Homogeneous mixture

A mixture in which different constituents are mixed uniformly, is called a homogeneous mixture. This mixture has same composition throughout. For example, salt solution, sugar solution. Similarly, alloys such as brass, bronze, etc. are homogeneous solid solutions of metals.

(b) Heterogeneous mixture

A mixture in which different constituents are not mixed uniformly, is called a heterogeneous mixture. For example, Sand and iron filings, sand and water, etc.

**Aim**

To illustrate the concept of homogeneous and heterogeneous mixtures.

Materials required

Beakers, water, spatula, copper sulphate powder, potassium permanganate or common salt

Method

- Let us divide the class into groups A, B, C and D.
- Group A takes a beaker containing 50 mL of water and one spatula full of copper sulphate powder. Group B takes 50 mL of water and two spatulas full of copper sulphate powder in a beaker.
- Group C and D can take different amounts of copper sulphate and potassium permanganate or common salt (sodium chloride) and mix the given components to form a mixture.

- Report the observations on the uniformity in colour and texture.

Observation and discussion

- Both groups A and B have obtained homogeneous mixtures since the composition of these mixtures or solutions is uniform throughout.



Hydrogen chloride gas is a pure substance since it is a chemical compound i.e. HCl(g) .

However, hydrochloric acid prepared by passing the gas into water is a homogeneous mixture. It is therefore, a solution.

SPOT LIGHT

- (ii) Although both the groups have obtained copper sulphate solutions but the intensity of colour of the two solutions is different. The intensity of blue colour in the solution obtained by group B which contains two spatulas full of copper sulphate is much higher than the solution obtained by group A which contains one spatula full of copper sulphate.
- (iii) Both groups C and D have obtained heterogeneous mixtures since they not only have physically distinct boundaries but also their composition is not uniform.

Conclusion

- (i) Soluble substances such as copper sulphate, common salt or sugar when dissolved in water form homogeneous mixtures, whose composition depend upon the amount of the substance dissolved.
- (ii) When two or more solids which do not react chemically are mixed, they always form heterogeneous mixtures.

In order to understand the difference between solution, colloid and suspension, let's perform the following activity.



Aim

To illustrate the difference between true solutions, suspensions and colloidal solutions.

Materials required

Beakers, glass rods, spatula, copper sulphate, chalk powder or wheat flour, milk or ink, torch, filtration apparatus.

Method

Let us again divide the class into four groups – A, B, C and D.

- (a) Distribute the following samples to each group.
- (i) Few crystals of copper sulphate to group A.
 - (ii) One spatula full of copper sulphate to group B.
 - (iii) Chalk powder or wheat flour to group C.
 - (iv) Few drops of milk or ink to group D.
- (b) Each group should add the given sample in water and stir properly using a glass rod. Are the particles in the mixture visible?
- (c) Direct a beam of light from a torch through the beaker containing the mixture and observe from the front. Was the path of the beam of light visible?
- (d) Leave the mixtures undisturbed for a few minutes (and set up the filtration apparatus in the meantime). Is the mixture stable or do the particles begin to settle after some time?
- (e) Filter the mixture. Is there any residue on the filter paper?

Observation and discussion**1. Group A and B**

- (i) Group A and B both have obtained true solutions. However, the intensity of the blue colour of solution obtained by group A which contains only a few crystals of copper sulphate is lower than that of the solution obtained by group B which contains one spatula full of copper sulphate. Thus, the intensity of colour depends upon the amount of substance dissolved.
- (ii) In both these solutions, particles are not visible to the naked eye.
- (iii) When the beam of light is passed through solution obtained by either group A or group B, the path of the beam of light was not visible. Thus, the particles of a true solution do not scatter light.
- (iv) When the above solution is allowed to stand, the solution is stable and the particles of the solution do not settle down. Further, when the above solution is filtered, the whole solution passes through the pores of the filter paper without leaving any residue on the filter paper.
Thus, the particles of a true solution neither settle down on standing nor leave any residue when passed through a filter paper.

2. Group C

- (i) The group C has obtained a suspension.
- (ii) The particles of a freshly made suspension scatter light and hence are visible to the naked eye.
- (iii) When the above suspension is allowed to stand, it is not stable and the particles of the suspension settle down.
- (iv) When the above suspension is filtered, the particles of the chalk or the wheat flour being bigger than the pores of the filter paper remain as residue on the filter paper.

3. Group D

- (i) The group D has obtained a colloidal solution.
- (ii) When the above solution is allowed to stand, the particles of the colloidal solution like those of the true solution do not settle down.
- (iii) Since the particles of a colloidal solution are about 100 times bigger than those of the true solution, therefore, when a beam of light is passed through a colloidal solution, the particles of the colloidal solution scatter light and hence the path of the beam becomes visible.
- (iv) When the colloidal solution is passed through a filter paper, the particles of the colloidal solution like those of the true solution pass through the filter paper without leaving any residue on the filter paper.

Conclusion

- (i) The particles of a true solution are not visible even under a microscope. They do not settle down on standing and pass through the pores of the filter paper without leaving any residue.
- (ii) The particles of a suspension are visible to the naked eye. They settle down on standing and leave a residue on the filter paper.
- (iii) The particles of a colloidal solution scatter light and hence the path of light beam becomes visible when a beam of light is passed through it. A colloidal solution is stable and hence the particles of the colloidal solution do not immediately settle down on standing. They also pass through the pores of a filter paper.



A solution in which water acts as the solvent is called an aqueous solution while the one in which any other liquid acts as the solvent is called a non-aqueous solution.

SPOT LIGHT**Check your
Concepts****1**

1. Identify pure substance and mixtures from the following.
Air, salt solution, milk, sand in water, water, nitrogen, sodium.
2. Identify homogeneous and heterogeneous mixtures from the following.
Air, salt solution, chalk in water, sand in water.

4. Solution

A solution may be defined as a homogeneous mixture of two or more non-reacting substances whose composition can be varied within certain limits.

For example, Air is a mixture of gas in gas. Air is a homogeneous mixture of a number of gases. Its two main constituents are oxygen (21%) and nitrogen (78%).

It may be noted that all mixtures are not solutions. If a mixture is to be called a solution, it must satisfy the following two conditions.

- Components should be non-reacting.
- Should be homogeneous.

Components of a solution

The substances present in a homogeneous solution, are called components of the solution. A solution basically has two components i.e. a solvent and a solute and such a solution is called a binary solution.



Solution of potassium iodide and iodine in water is known as lugol.

(a) Solute:

The component of the solution which dissolves in the solvent, is called solute. Usually, solute is the smaller component of the solution, for example, solution of iodine in alcohol, known as 'tincture of iodine', iodine is the solute in this solution.

Similarly, in carbonated drinks (soda water), carbon dioxide gas is the solute.

(b) Solvent

The component of a solution which dissolves the other component in itself, is called solvent. Usually, a solvent is the larger component of the solution. For example, a solution of sugar in water is a solid in liquid solution. In this solution, sugar is the solute and water is the solvent.

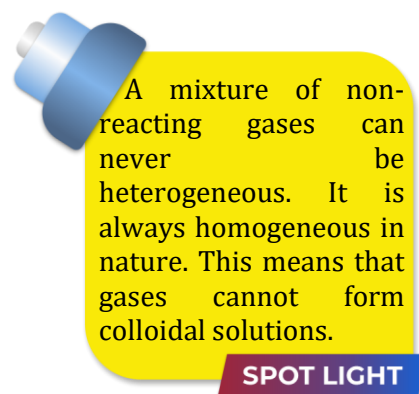
The common examples of aqueous solutions are

- (i) common salt dissolved in water.
- (ii) sugar dissolved in water.
- (iii) acetic acid dissolved in water (called vinegar).

The common non-aqueous solvents are; alcohol, carbon disulphide, carbon tetrachloride, acetone, benzene, etc.

Examples of non-aqueous solutions are

- (i) Iodine dissolved in carbon tetrachloride.
- (ii) Sulphur dissolved in carbon disulphide.
- (iii) Bromine dissolved in chloroform.
- (iv) Sugar dissolved in alcohol, etc.

**Characteristics of a solution**

- (1) A solution is a homogeneous mixture.
- (2) The size of solute particles in a solution is extremely small. It is less than 1 nm (10^{-9} m) in diameter.
- (3) The particles of a solution cannot be seen even with a microscope.
- (4) The particles of a solution pass through the filter paper. So, a solution cannot be separated by filtration.
- (5) The solutions are very stable. The particles of solute present in a solution do not separate out on keeping it undisturbed.
- (6) A true solution does not scatter light passing through the solution (because its particles are very small).
- (7) The components of a solution do not chemically react with one another.
- (8) A solution is always transparent in nature.
- (9) From a true solution, the solute can be easily recovered by evaporation or crystallisation.

Methods to express concentration of solution

The concentration of a solution is the amount of solute present in a given quantity of the solution or solvent.

In a solution the relative proportion of the solute and solvent can be varied. Depending upon the amount of solute present in a solution, it can be called a dilute, concentrated or a saturated solution.

There are many ways to express the concentration of a solution. But the most common method for expressing the concentration of a solution, is called percentage method.

(i) Mass by mass percentage(w/w)

The mass of the solute in grams dissolved in 100 g of the solution.

$$\text{Mass by mass percentage of a solution} = \frac{\text{Mass of solute (in g)}}{\text{Mass of solution (in g)}} \times 100$$

(ii) Mass by volume percentage(w/v)

The mass of the solute in grams dissolved in 100 mL of the solution

$$\text{Mass by volume percentage} = \frac{\text{Mass of solute (g)}}{\text{Volume of solution (ml)}} \times 100$$

(iii) Volume by volume percentage or simply volume percentage(v/v).

The volume of the solute in millilitres dissolved in 100 mL of the solution.

$$\text{Volume by volume percentage} = \frac{\text{Volume of solute (ml)}}{\text{Volume of solution (ml)}} \times 100$$



★ Mass of solution = Mass of solute + Mass of solvent



A solution contains 40 g of common salt in 320 g of water. Calculate the concentration in terms of mass by mass percentage of the solution.

Solution:

Mass of solute (salt) = 40 g

Mass of solvent (water) = 320 g

We know,

Mass of solution = Mass of solute + Mass of solvent = 40 g + 320 g = 360 g

$$\text{Mass percentage of solution} = \frac{40}{360} \times 100\% = 11.1\%$$



How much water should be added to 50 g of glucose, so as to obtain 12% glucose solution?

Solution:

Percentage concentration of glucose solution = 12%

Mass of solute = 50 g

Mass of solution = ?

$$\text{Concentration of solution} = \frac{\text{Mass of solute (g)}}{\text{Mass of solution}} \times 100$$

$$\therefore \text{Mass of solution} = \frac{\text{Mass of solute (g)}}{\text{Concentration of solution}} \times 100 = 416.67 \text{ g}$$

$\therefore$ Mass of water which should be added to glucose

$$= \text{Mass of solution} - \text{Mass of solute} = (416.67 - 50) \text{ g} = 366.67 \text{ g}$$



A 250 mL of acetic acid solution in water contains 20 g of acetic acid. What is the concentration of acetic acid solution?

Solution:

Mass of solute = 20 g

Volume of solution = 250 mL

Mass by volume percentage

$$= \frac{\text{Mass of solute}}{\text{Volume of solution}} \times 100 = \frac{20\text{g}}{250\text{mL}} \times 100 = 8\%$$

Classification of solutions on the basis of amount of solute



Aim

To demonstrate that different substances have different solubilities in the same solvent.

Materials required

Beakers, water, sugar, glass rod, barium chloride, common salt.

Method

- Take approximately 50 mL of water each in two separate beakers.
- Add common salt in one beaker and sugar or barium chloride in the second beaker with continuous stirring.
- When no more solute can be dissolved, heat the contents of the beaker.

- (d) Start adding the solute again.
- (e) Is the amount of common salt and sugar or barium chloride, that can be dissolved in water at a given temperature, the same?
- (f) What would happen if you were to take a saturated solution at a certain temperature and cool it slowly?

Discussion

- (i) The amounts of common salt, sugar and barium chloride that can be dissolved in water (50 mL) at room temperature are different.
- (ii) When a saturated solution at a certain temperature is cooled, the solubility decreases and the amount of the solute which exceeds the solubility at the lower temperature crystallizes out of the solution.

Conclusion

Different substances have different solubilities in a given solvent at the same temperature and in general the solubility decreases as the solution is cooled and the extra amount of the solute crystallizes out.

Saturated solution

A solution which at a given temperature dissolves as much solute as it is capable of dissolving is said to be a saturated solution.

Example:

- (a) When we add sugar to water, the crystals of sugar keep on going into the solution in the beginning. But after some time, no mass of sugar dissolves, provided temperature is kept constant.
- (b) At 30°C, 55 g of common salt dissolves in 100 g of water. However, if more of common salt is added to the above solution, it does not dissolve. In such a situation, the solution of common salt containing 55 g of salt in 100 g of water is a saturated solution at 30°C.
- If a saturated solution at some particular temperature is heated, the solution becomes unsaturated, because of the increase in solubility.
 - If a saturated solution at some higher temperature is cooled, it remains saturated. The excess solute comes out of the solution and deposits itself in the form of crystals.

Solubility

The maximum amount of solute in grams which can be dissolved in 100 g of the solvent at a given temperature to form a saturated solution is called solubility of the solute in that solvent at that particular temperature.

Unsaturated solution

When the amount of solute contained in a solution is less than the saturation level, the solution is said to be an unsaturated solution.

For example, At 30°C, if 45 g of common salt is dissolved in 100 g of water, such solution so formed is capable of dissolving more of the common salt, then such a solution is called an unsaturated solution.

Super saturated solution

A solution which contains more of the solute than required to make a saturated solution, is called a super saturated solution.

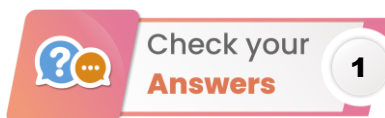
Solubility of some substances at 20°C

Substance (or Solute)		Solubility in water (at 20°C)
1.	Copper sulphate	21 g
2.	Potassium nitrate	32 g
3.	Potassium chloride	34 g
4.	Sodium chloride	36 g
5.	Ammonium chloride	37 g
6.	Sugar	204 g

**How can we convert a saturated solution into an unsaturated solution?****Explanation**

There are two ways to convert a saturated solution into unsaturated:

- By increasing the temperature: Because on heating, solubility of the solute increases and hence the solution becomes unsaturated.
- By adding more solvent.



- Air, salt solution, milk, sand in water - Mixtures
Water, nitrogen, sodium - Pure substances
- Air, salt solution - Homogeneous mixture
Chalk in water, sand in water - Heterogeneous mixture

5. Suspension

A suspension is a heterogeneous mixture in which the solid particles are spread throughout the liquid without dissolving in it. The particles have a tendency to settle down at the bottom of the container and can be filtered out.

For example,

(1) Chalk-water mixture is a suspension of fine chalk particles in water.

(2) Muddy water is a suspension of soil particles in water.

Properties of a suspension

(1) A suspension is a heterogeneous mixture.

(2) The size of solute particles in a suspension is quite large. It is larger than 1000 nm or 10^{-6} m in diameter.

(3) The particles of a suspension can be seen easily with naked eyes and under microscope.

(4) A suspension scatters a beam of light passing through it (because its particles are quite large, thus makes its path visible).



Pearls are solid sol

(5) The particles of suspension settle down when the suspension is kept undisturbed.

(6) The process of settling down of suspended particles under the action of gravity is called sedimentation. So, suspensions are unstable. After sedimentation, it does not scatter light any more.

(7) The solid particles present in the suspension can be easily separated by ordinary filter papers. No special filter paper is needed for the purpose.

(8) Suspensions are either opaque or translucent.

6. Colloidal solution or colloids

A heterogeneous solution in which the particle size is in between 10^{-7} cm to 10^{-4} cm such that the solute particles neither dissolve nor settle down in a solvent, is called colloidal solution. Blood is an example of colloid. Its components can be separated using centrifugation.



Smoke is an example of solid aerosol

The components of a colloidal solutions are the dispersed phase and the dispersion medium.

Dispersed phase

It is the component which is present in small proportion and consists of particles of colloidal dimensions (10^{-9} m to 10^{-6} m).

Dispersion medium

It is the component in which dispersed phase is suspended and acts as a medium.

Colloids are classified according to the state of the dispersing medium.

Types of colloids			
Dispersed Phase	Dispersion medium	Type of colloid	Examples
Liquid	Gas	Liquid Aerosol	Mist, Fog, clouds, insecticide sprays
Solid	Gas	Solid Aerosol	Smoke, dust in air, smog, automobile exhaust
Gas	Liquid	Foam, Froth	Whipped cream, froth, shaving cream
Liquid	Liquid	Emulsion	Milk, hair cream, emulsified oils, medicines, cold cream
Solid	Liquid	Sol	Paints, starch, proteins, milk of magnesia, mud, blood
Gas	Solid	Solid foam	Foam, pumice stone, ice-cream, rubber, sponge
Liquid	Solid	Gel	Jellies, cheese, butter, boot polish
Solid	Solid	Solid sol	Coloured glass, gemstones, pearls, milky glass

Properties of colloids

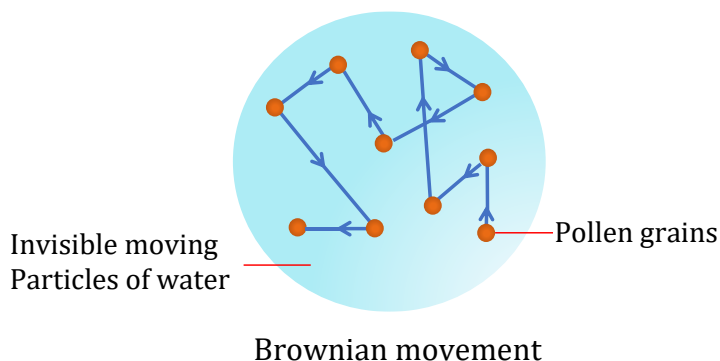
- (1) A colloid is a heterogeneous mixture but it appears homogeneous.
- (2) The size of particles of a colloid is too small to be individually seen by naked eyes.
- (3) The colloidal particles do not settle down when left undisturbed i.e. colloid is quite stable.
- (4) The colloid particles cannot be separated from the dispersion medium by the process of filtration. A special technique of separation known as centrifugation can be used to separate the colloidal particles.
- (5) Colloidal solutions are not transparent, but translucent in nature.
- (6) Particles in a colloidal solution follow zigzag path (Brownian movement).



The fog you usually observe during winters is an example of liquid aerosol

It is normally not possible to see the colloidal particle because of their very small size. However, their path can be seen under a microscope.

Brownian movement: It is defined as continuous zig-zag movement of colloidal particles in a colloid. This Brownian movement arises due to hitting of the colloidal particles by the particles of the dispersion medium from different directions with different forces.



Colloid particles follow a zig-zag path. Such type of movement was noticed for the first time by Robert Brown, an English scientist, hence is called Brownian movement after his name.

SPOT LIGHT

(7) Colloidal solution scatters the beam of light passing through it. **(Tyndall effect)**

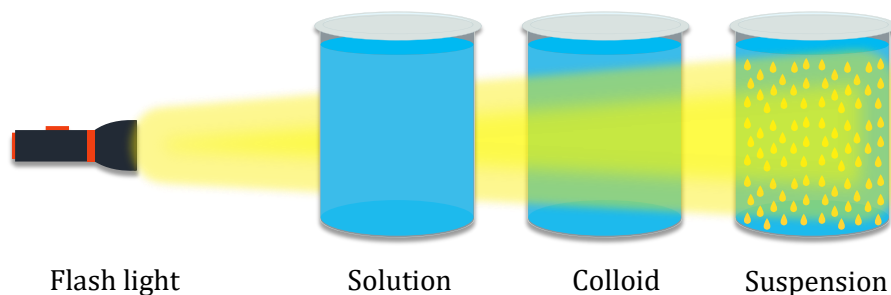
As a result, the path of light becomes visible. **This scattering of a beam of light by colloidal particles is called the Tyndall effect.** The reason for this observation is that the particles of a colloidal solution are big enough to scatter light and hence path of light becomes visible i.e. the Tyndall effect is observed.

For example,

- (i) Tyndall effect can also be observed when a fine beam of light enters a room through a small hole. This happens due to the scattering of light by the particles of dust and smoke in the air.
- (ii) Tyndall effect can be observed when sunlight passes through the canopy of a dense forest. In the forest, mist contains tiny droplets of water, which acts as particles of colloids dispersed in air.



Tyndall effect



Tyndall effect


**Quick
Tips**

- ★ Difference between true solutions, suspensions and colloidal solutions

Property	True Solutions	Suspension	Colloidal
Particle size	Less than 10^{-9} m/ 10^{-7} cm/1 nm	Greater than 10^{-6} m/ 10^{-4} cm / 1000 nm	Between (10^{-9} to 10^{-6} m)/ (10^{-7} to 10^{-4} cm)/ (1 to 1000 nm)
Visibility of particles	Invisible to naked eye and under powerful microscope.	Easily visible	Invisible to naked eye. Visible under powerful microscope.
Sedimentation of particles	Do not settle down.	Settle down due to gravity.	Settle down under high centrifugation.
Filtration through filter paper	No residue is formed.	Residue is formed.	No residue is formed.
Mixture type	Homogeneous	Heterogeneous	Heterogeneous
Appearance	Clear and Transparent	Translucent or Opaque	Translucent


**Check your
Concepts**

2

- Out of a colloid, solution and a suspension
(a) which one has the smallest particles?
(b) which one has the largest particles?
- Which of the two will scatter light: soap solution or sugar solution? Why?
- Which of the following will show Tyndall Effect? Why?
(a) Salt solution (b) Starch solution (c) Milk (d) Copper sulphate solution



A special technique of separation known as centrifugation, can be used to separate the colloidal particles.

SPOT LIGHT


**Check your
Answers**

2

- (a) Solution (b) Suspension
- Soap solution, because it is a colloid.
- (a) No, as it is a true solution. (b) Yes, as it is colloid in nature.
(c) Yes, colloid (d) No, true solution

7. Physical and chemical changes

There are some changes during which no new substances are formed. On the other hand, there are some other changes during which new substances are formed. So, on the basis of whether new substances are formed or not, we can classify all the changes into two groups:

- (i) Physical changes
- (ii) Chemical changes

Physical changes

Those changes in which no new substances are formed, are called physical changes. In a physical change, the substances involved do not change their identity. The changes in physical state, size and shape of substances are called physical changes.

When ice is heated, it melts to form water. Though ice and water look different, they are both made of water molecules. Thus, no new chemical substance is formed during the melting of ice. So, the melting of ice to form water is a physical change.

When water is cooled, it solidifies to form ice. This is called freezing of water. The freezing of water to form ice is also a physical change.

Some other example of physical changes are: Boiling of water, condensation of steam, and breaking of a glass.

Chemical changes

Those changes in which new substances are formed, are called chemical changes.

In a chemical change, the substances involved change their identity. They get converted into entirely new substances.

For example,

When a magnesium wire is heated, it burns in air to form a white powder called 'magnesium oxide'. This magnesium oxide is an entirely new substance. Thus, a new chemical substance is formed during the burning of a magnesium wire. This is a chemical change.

Some other examples of chemical changes are: Burning of candle and burning of hydrogen in oxygen to form water.



When water boils no new substance is formed, so boiling is a physical change



Cutting of trees is an irreversible process but no new substance is formed during cutting of trees. Therefore, we call it a physical change

Differences between physical and chemical changes	
Physical change	Chemical change
A physical change brings about changes in physical properties of the substance such as physical state, shape and size, etc. For example, ice (solid) melts to form water (liquid).	A chemical change brings about changes in the chemical properties of the substance. For example, iron undergoes rusting to form hydrated iron oxide. The chemical properties of hydrated iron oxide are different from those of iron.
There is no change in the chemical composition of the substance during the physical change. Thus, both ice and steam have the same chemical composition (made up of water molecules) as that of liquid water.	There is always a change in the chemical composition of a substance during a chemical change. For example, iron and rust have different chemical composition.
No new substance is formed in a physical change.	A new substance is always formed during a chemical change.



Can physical and chemical change take place together? Give examples.

Explanation

Yes, burning of candle. The solid wax present in the candle first changes into liquid state and then into the vapour state. Both these changes are physical changes. Then the wax vapour combine with oxygen of air to form a mixture of CO_2 and H_2O . This is a chemical change.



Formation of fruit salad is another example of physical change

8. Types of pure substances

On the basis of their chemical composition, substances can be classified either as elements or compounds.

Element

Antoine Laurent Lavoisier defined element as the simplest or basic form of matter that cannot be broken into simpler substances by chemical methods.

OR

A pure substance which is made up of one kind of atoms only is called an element.

Majority of the elements are solid. Eleven elements are in gaseous state at room temperature.

Two elements are liquid at room temperature, mercury and bromine.

Types of elements**(1) Metals**

Nearly 70 elements belong to a particular class known as metals. In the metals, the atoms are very closely packed together and have special types of bonds known as metallic bond. Because of very tight packing, the metals are quite hard. Cesium and gallium are two elements which liquify slightly above room temperature.

(2) Non-metals

Nonmetals as the name suggests are opposite to metals which means that their properties are quite different from the metals. Only about 22 elements are nonmetals.

Properties of metals

- (1) They have generally silver-grey colour. However, some metal or their alloys have golden yellow colour.
- (2) Metals have a lustre, the freshly cut surface has a shine on it.
- (3) They easily conduct heat and electricity.
- (4) They are malleable i.e. they can be beaten into sheets.
- (5) They are sonorous.

For example, Gold, silver, copper, iron, sodium, potassium, etc. Mercury is the only metal that is liquid at room temperature.

Properties of non-metals

- (1) They exist in solid, liquid and gaseous state.
- (2) They display variety of colour.
- (3) They are generally neither malleable nor ductile.
- (4) They are poor conductors of heat and electricity.
- (5) They are not sonorous.

For example, Hydrogen, oxygen, iodine, carbon, etc.

Metalloids

Some element have intermediate properties of the metals and nonmetals. The elements which exhibit the properties of metals as well as nonmetals are called metalloids. For example, Boron, Silicon, Germanium, etc.



Robert Boyle was the first scientist to use the term "Element" in 1661

Compounds

A pure substance, which is composed by two or more elements, combined chemically in a definite ratio by mass, such that it can be broken into elements only by chemical means, is called compound. The two or more elements present in a compound, are called constituents or components of the compound. For example, water is a compound of hydrogen and oxygen, combined together in the ratio of 1 : 8 by weight. The water can be broken into its constituents only by electrochemical method i.e. by passing electric current through it. The compounds can be further classified as acids, bases and salts.

Difference between mixtures and compounds	
Mixtures	Compounds
Elements or compounds just mix together to form a mixture and no new compound is formed.	Elements react to form new compounds.
A mixture has variable composition.	The composition of each new substance is always fixed.
A mixture shows the properties of the constituent substances.	The new substance has totally different properties.
The constituents can be separated easily by physical methods.	The constituents can be separated only by chemical or electrochemical methods.



Aim

To study the properties of mixtures and compounds.

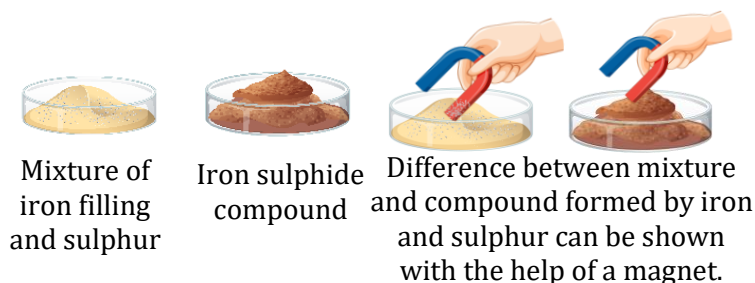
Materials required

China dish, iron filings, sulphur powder, burner, magnet, beaker sulphuric acid or dilute hydrochloric acid.

Method

- Divide the class into two groups. Give 5 g iron filings and 3 g of sulphur powder in a china dish to both the groups.
- Group-I:** Mix and crush iron filing and sulphur powder.
- Group-II:** Mix and crush iron filings and sulphur powder. Heat this mixture strongly till red hot. Remove the flame and let the mixture cool.
- Group-I and II:** Check for magnetism in the material obtained. Bring a magnet near the material and check if the material is attracted towards the magnet.

- (e) Compare the texture and colour of the material obtained by the groups. Add carbon disulphide to one part of the material obtained. Stir well and filter.
- (f) Add dilute sulphuric acid or dilute hydrochloric acid to the other part of the material obtained.

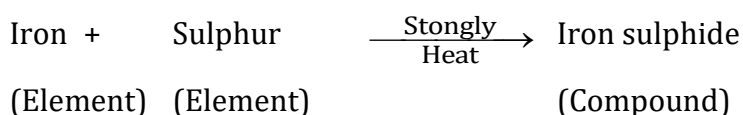


Now answer

- (a) Did the material obtained by the two groups look the same?
- (b) Which group has obtained a material with magnetic properties?
- (c) Can we separate the components of the material obtained?
- (d) On adding dilute sulphuric acid or dilute hydrochloric acid, did both the groups obtain a gas? Did the gas in both cases smell the same or different?

Discussion

- (a) No,
- (i) The material obtained by students of group I is a heterogeneous mixture, i.e. when we examine the mixture under a microscope, it is found that though the iron particles lie very close to the sulphur particles, at some places there are more of iron particles whereas at other places, there are more of sulphur particles.
- (ii) On the other hand, the material obtained by students of group II is a compound which has altogether different properties as compared to those of its constituents i.e. iron filings and sulphur.



- (iii) If we examine iron sulphide under a microscope, we find that the composition, texture and colour of the compound (black) is the same throughout.
- (b) Material obtained by group I contains iron filings which have magnetic properties. Alternatively, material obtained by group II is a compound which does not contain iron filings and hence does not have magnetic properties.
- (c) The material obtained by students of group I is a mixture and hence its components can be separated by simple physical methods such as a magnet or carbon disulphide.

If we put a magnet in the mixture of iron filings and sulphur powder, the iron particles are attracted by the magnet, they stick to the magnet and get separated from sulphur. Alternatively, if we shake the mixture of iron filings and sulphur powder with carbon disulphide (CS_2), sulphur dissolves in it but iron filings do not. On filtration, iron filings are obtained as a residue on the filter paper while sulphur is recovered from the filtrate by evaporating carbon disulphide.

- (d) Material obtained by students of both group I and group II on treatment with dilute sulphuric acid or dilute hydrochloric acid evolve a gas but the gas evolved is different in each case.

When dilute sulphuric acid is added to the material (mixture of iron filings and sulphur powder) obtained by students of group I, the iron part reacts with dilute sulphuric acid to produce *hydrogen gas* (H_2) which is a colourless, odourless and combustible gas. When dilute sulphuric acid is added to the material (iron sulphide) obtained by students of group II, *hydrogen sulphide gas* (H_2S) is obtained.

It is a colourless gas having the smell of rotten eggs.



Purity of compounds can be tested by determining their melting or boiling points. Pure compounds have fixed melting point and boiling point.

SPOT LIGHT

Conclusion

A mixture shows the properties of its constituents, but the properties of a compound are altogether different from those of its constituents.

Basic terminology

1. **Transparent** - You can see through it clearly.
2. **Translucent** - You can see through it partially.
3. **Opaque** - You cannot see through it.
4. **Volatile** - Substance that forms vapours on heating.
5. **Non-volatile** - Substance that does not form vapours on heating.
6. **Condenser** - The part of apparatus where condensation takes place.
7. **Charring** - Process of heating of sugar to form carbon black.
8. **Alum** - Fitkari.
9. **Hydrocarbon** - A compound of hydrogen and carbon.
10. **Constituent** - A component part of something.
11. **Uniformly** - In a way that is the same in all cases.
12. **Homogeneous** - Of the same kind; Alike
13. **Heterogeneous** - Diverse in character or content.
14. **Composition** - Arrangement, ratio, and type of atoms in molecules.
15. **Scatter** - Throw in various random directions.
16. **Pore** - A minute opening in a surface.

17. **Tendency** - Typical or repeated habit, action or belief.
18. **Canopy** - A projection or shelter that resembles a roof.
19. **Smoke** - A visible suspension of carbon or other particles in air.
20. **Smog** - A mixture of smoke, gases and chemicals.
21. **Mist** - Thin layer produced by very small droplets of water collecting in the air.
22. **Fog** - A thick cloud of tiny droplets suspended in air.
23. **Inflammability** - Capable of being set on fire
24. **Extinguish** - Put an end to
25. **Rotten** - Very bad

Memory Map

